

Lucas Narloch Zabla

Software Engineer | Backend & Cloud Systems

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Computer Science student and Backend Developer with nearly two years of experience building and operating production software on AWS. Experienced with Python, distributed and event-driven systems, SQL and NoSQL databases, automated testing, CI/CD, infrastructure as code, observability, and production troubleshooting. Hands-on computer science foundations through compiler and systems programming projects in C.

TECHNICAL SKILLS

- **Languages:** Python, C, TypeScript, SQL
- **CS Fundamentals:** Data Structures & Algorithms, Complexity Analysis, Object-Oriented Programming, Operating Systems, Computer Networks
- **Backend & Distributed Systems:** REST APIs, Event-Driven Architecture, Amazon SQS, AWS Lambda, MQTT
- **Databases:** PostgreSQL, MySQL, DynamoDB
- **Cloud & DevOps:** AWS, Docker, Terraform, GitHub Actions, CI/CD, Linux, Bash, Git
- **Testing & Reliability:** pytest, Testcontainers, Unit Testing, Integration Testing, Observability, Production Troubleshooting

EXPERIENCE

Backend Developer Intern - SmartLy Fabricação de Dispositivos Inteligentes Ltda.

09/2024 - Present

- Develop and operate Python backend services and REST APIs integrated with AWS, SQL and NoSQL databases, and IoT systems.
- Diagnosed production failures using structured logs, metrics, and AWS CloudWatch, optimized Python Lambda workloads, and reduced errors from 113 to 5 per five-minute interval (95.6%).
- Designed and implemented asynchronous, event-driven workflows with Amazon SQS, handling peaks above 40,000 notifications per day while decoupling background processing from latency-sensitive API requests.
- Built an IoT data migration endpoint across MySQL, DynamoDB, S3, and IoT Core via MQTT, supporting thousands of historical records per operation.
- Built 30 integration tests and 80 unit tests with pytest and Testcontainers, reducing production regression risk.
- Contributed to consolidating AWS Lambda functions into a structured FastAPI application, centralizing business rules and reducing operational complexity.
- Automated infrastructure, deployments, and environment configuration with Terraform, GitHub Actions, Docker, and Bash, reducing manual work and configuration risk.
- Collaborate across backend, frontend, firmware, and product teams on architecture, observability, and production reliability.

PROJECTS

PocScript

Programming language and compiler implemented in C, covering lexical analysis, AST parsing, semantic analysis, custom intermediate representation, LLVM IR emission, and native executable generation. Includes a Linux syscall-based runtime, static type system, scope resolution, arrays, pointers, and 196 unit and integration tests.

Technologies: C, LLVM IR, NASM, Make, Linux system calls, Compiler Design

[Code](#)

Sunlog.dev

Production web platform with 33 REST endpoints, PostgreSQL, JWT authentication, asynchronous processing, and more than 100 automated tests. Deployed on AWS using Docker and CI/CD pipelines, with structured logging and monitoring through Prometheus and Grafana.

Technologies: NestJS, TypeScript, PostgreSQL, Docker, AWS, GitHub Actions, Prometheus, Grafana

[Code](#) | [Live demo](#)

EDUCATION

Bachelor of Computer Science - PUCPR

02/2025 - 12/2028

Full scholarship recipient.

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Databases, Software Engineering

Technical High School Diploma in Electronics - CEEP

02/2022 - 12/2024

LANGUAGES

- **Portuguese:** Native
- **English:** Professional Working Proficiency